

Finite Difference Gradient and Hessian for Newton's Method

Finite Difference Gradient (Central Difference)

For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the i -th component of the gradient can be approximated using the central difference formula:

$$\frac{\partial f(x)}{\partial x_i} \approx \frac{f(x + he_i) - f(x - he_i)}{2h}$$

where

$$e_i = [0, \dots, 0, 1, 0, \dots, 0]^T$$

is the unit vector in the i -th coordinate direction and h is a small step size.
Thus, the gradient vector is approximated as

$$\nabla f(x) = \begin{bmatrix} \frac{f(x + he_1) - f(x - he_1)}{2h} \\ \frac{f(x + he_2) - f(x - he_2)}{2h} \\ \vdots \\ \frac{f(x + he_n) - f(x - he_n)}{2h} \end{bmatrix}$$

Finite Difference Hessian

The Hessian matrix can be approximated using second-order central differences:

$$H_{ij}(x) \approx \frac{f(x + he_i + he_j) - f(x + he_i - he_j) - f(x - he_i + he_j) + f(x - he_i - he_j)}{4h^2}$$

The Hessian matrix therefore becomes

$$H(x) = \begin{bmatrix} H_{11} & H_{12} & \dots & H_{1n} \\ H_{21} & H_{22} & \dots & H_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ H_{n1} & H_{n2} & \dots & H_{nn} \end{bmatrix}$$

Newton Update Step

Using the gradient and Hessian, the Newton search direction is

$$p_t = H(x_t)^{-1} \nabla f(x_t)$$

and the update step is

$$x_{t+1} = x_t - \alpha p_t$$

where α is a scaling factor or step size.